Activity 6.1: Arcs, Angles & Segments in Circles

A

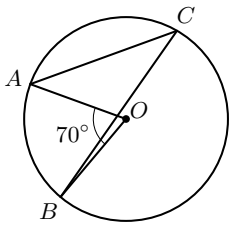
Name: _____ Partner's name: _____ Date: _____

Directions: Find the value or measure asked for. Give exact answers in terms of π when needed. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

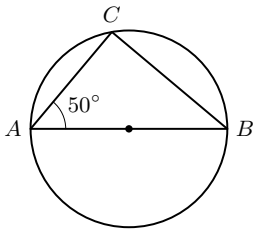
The central angle $\angle AOB$ measures $70°$. Find $m\angle ACB$.



Answer: _____ Match? ☐ Yes ☐ No

2

$\overline{AB}$ is a diameter and $m\angle A = 50°$. Find $m\angle B$.



Answer: _____ Match? ☐ Yes ☐ No

3

Find the length of an arc with a central angle of $60°$ in a circle of radius 6.

Answer: _____ Match? ☐ Yes ☐ No

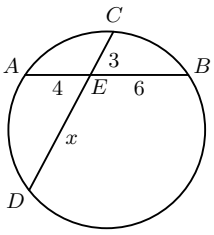
4

Find the area of a sector with a central angle of $60°$ in a circle of radius 6.

Answer: _____ Match? ☐ Yes ☐ No

5

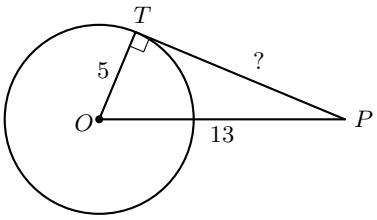
Chords $\overline{AB}$ and $\overline{CD}$ intersect at E. Find x.



Answer: _____ Match? ☐ Yes ☐ No

6

$\overline{PT}$ is tangent to the circle at T. Find PT.



Answer: _____ Match? ☐ Yes ☐ No

7

Two secants intersect outside a circle and intercept arcs of $100°$ and $40°$. Find the measure of the angle they form.

Answer: _____ Match? ☐ Yes ☐ No

8

Two chords intersect inside a circle. The angle formed intercepts arcs of $70°$ and $50°$. Find the measure of that angle.

Answer: _____ Match? ☐ Yes ☐ No

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B

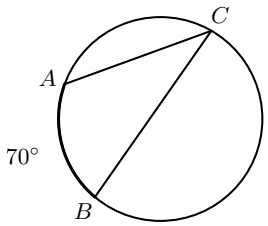
Name: _____ Partner's name: _____ Date: _____

Directions: Find the value or measure asked for. Give exact answers in terms of π when needed. Diagrams are accurate, but use math rather than a ruler or protractor.

Work alone, then compare after each problem. If answers differ, find the mistake together.

1

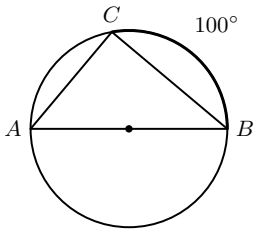
Minor arc AB measures $70°$. Find $m\angle ACB$.



Answer: _____ Match? ☐ Yes ☐ No

2

$\overline{AB}$ is a diameter and arc BC measures $100°$. Find $m\angle B$.



Answer: _____ Match? ☐ Yes ☐ No

3

Find the length of an arc with a central angle of $30°$ in a circle of radius 12.

Answer: _____ Match? ☐ Yes ☐ No

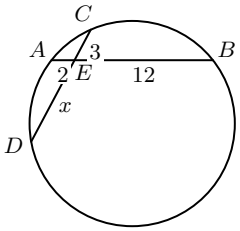
4

Find the area of a sector with a central angle of $15°$ in a circle of radius 12.

Answer: _____ Match? ☐ Yes ☐ No

5

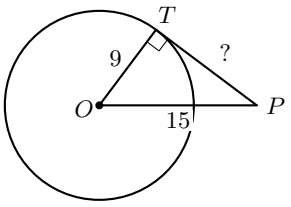
Chords $\overline{AB}$ and $\overline{CD}$ intersect at E. Find x.



Answer: _____ Match? ☐ Yes ☐ No

6

$\overline{PT}$ is tangent to the circle at T. Find PT.



Answer: _____ Match? ☐ Yes ☐ No

7

Two secants intersect outside a circle and intercept arcs of $110°$ and $50°$. Find the measure of the angle they form.

Answer: _____ Match? ☐ Yes ☐ No

8

Two chords intersect inside a circle. The angle formed intercepts arcs of $80°$ and $40°$. Find the measure of that angle.

Answer: _____ Match? ☐ Yes ☐ No